

A Live Demo of Single-Photon Imaging and Applications

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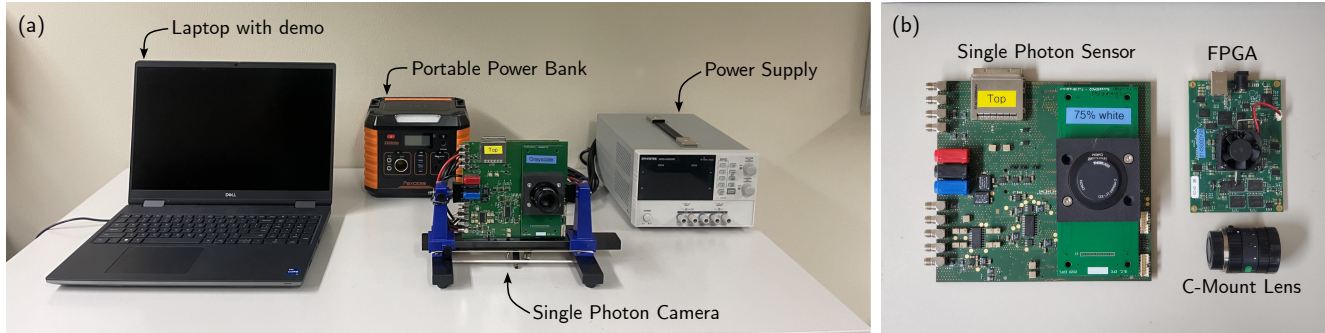


Figure 1: SwissSPAD2 imaging demo. (left) Our demo hardware consists of a 512×512 single-photon camera that interfaces with a host computer. **(right)** The SPAD prototype interfaces with a FPGA board for readout and can be fitted with C-mount lenses.

ABSTRACT

Single-photon sensors are a novel class of imaging sensors that are sensitive to the individual arrival of photons. In recent years, single-photon sensors have witnessed rapid growth in key sensor characteristics such as array resolution, which, for the first time, has made computer vision and imaging with these emerging sensors possible. This rapid growth has also spurred the development of algorithms that can harness the rich spatio-temporal scene information captured by single-photon sensors, which has led to several new imaging capabilities. These capabilities include imaging in challenging SNR conditions, high-dynamic range imaging, compensating extreme camera motion extents, and even providing the functionality of numerous imaging systems via post-capture processing. Our demo will showcase these exciting capabilities to a wide computer vision and graphics audience, and in doing so, make a case for the mainstream adoption of single-photon technology.

CCS CONCEPTS

• Computing methodologies → Computational photography.

KEYWORDS

Single-photon camera, single-photon avalanche diode, quanta image sensors, high dynamic range, high-speed imaging, low-light imaging, motion compensation, software-defined cameras.

*Denotes equal contribution.

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1 INTRODUCTION

Single-photon sensors are an emerging imaging technology that is sensitive to light at the granularity of individual photons. Notably, single-photon sensors, such as single-photon avalanche diodes (SPADs), can be operated as high-speed photon detectors (at speeds of up to ~ 100 kHz), producing a temporal sequence of binary frames that indicates the spatiotemporal detection of photons in each exposure window. The fine temporal resolution of single-photon acquisition yields several exciting imaging capabilities—including imaging in scenes with fast motion or low light [Ma et al. 2020], high-dynamic range imaging, motion compensation at very high speeds [Jungerman et al. 2023], and the emulation of a wide variety of imaging modalities *after the fact* [Sundar et al. 2023].

These algorithmic advances and new computer vision capabilities have in part been fueled by an exciting hardware trend—the emergence of high spatial-resolution SPAD arrays. Over the last decade, SPADs have witnessed rapid growth: from 32×32 pixels to recent megapixel SPADs [Morimoto et al. 2021]. Furthermore, owing to their compatibility with CMOS fabrication processes, SPADs are projected to become more mainstream and accessible, an early indication of which is the inclusion of dToF SPADs in newer iPhones. Bearing this hardware trend in mind, we believe now might be the right time to demonstrate the unique capabilities of SPADs to a wider vision and graphics audience.

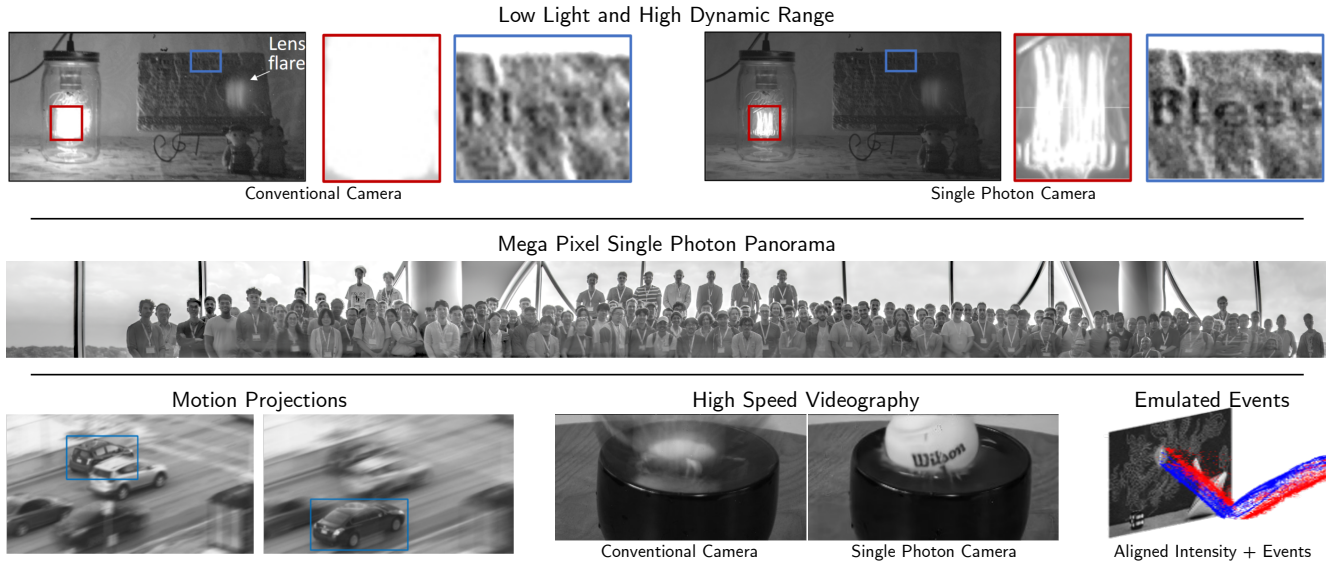


Figure 2: Novel capabilities of single-photon sensors. (top row) A single-photon camera, unlike a conventional image sensor, has a soft-saturating response to incident flux, resulting in an extreme dynamic range (200× here). (middle row) High-speed single-photon acquisition makes it possible to create panoramas from rapid handheld motion. Here, we show a 5.3 MPixel panorama captured at ICCP 2023. (bottom row) Single-photon data can also be transformed *after the fact* to emulate a variety of imaging modalities—including cameras that simulate camera motion, high-speed compressive captures, and event cameras.

2 HARDWARE DESCRIPTION

SPAD sensor. Our demo will utilize the SwissSPAD2 array [Ulku et al. 2019], which has a resolution of 512×256 pixels and a pixel pitch of $16.4\mu\text{m}$. The SwissSPAD2 is capable of reading out 1-bit images at 97.7 kHz for short durations (around 2 seconds) and multi-bit photon detection counts (which are summed binary frames) continuously for extended durations.

Sensor interface and power connections. As seen in Fig. 1 (left), the SwissSPAD2 camera module connects to an external power supply that delivers 29.5 V at 0.2 A and interfaces with a host computer using USB 3.0. The camera module hosts the SPAD sensor and connects with an FPGA board for sensor readout (shown in Fig. 1 (right)). Sensor readout is transferred to the computer, where we implement various single-photon algorithms. For an engaging experience, we will ensure that the overall acquisition and processing pipelines have latencies that are within real-time limits.

3 CAPABILITIES

We now detail imaging capabilities that we will demonstrate with the SwissSPAD2 array and their anticipated demo outcomes.

High-dynamic range imaging. A SPAD is capable of imaging both low-light regions (owing to single-photon sensitivity) and regions with excessive illumination simultaneously (due to a soft-saturating flux response, shown in Fig. 2 (top row)). This dynamic range is particularly evident when imaging a static scene since very long exposures can be taken. We will demonstrate this capability of single-photon sensors by imaging a specular object (a shiny metallic ball) placed in a dark box. For comparison, we will also capture the same scene with a consumer-grade DSLR camera.

Capturing a panorama with rapid handheld motion. The high-speed nature of a single-photon camera enables reliable motion estimation even under rapid camera movements [Jungerman et al. 2023]. As an application of high-speed motion estimation, we will demonstrate panorama stitching from SPAD data which we will acquire by having an audience member swivel a freestanding SwissSPAD2. Fig. 2 (middle) shows an example panorama mosaic that we captured at ICCP 2023 which was held in June at Madison, WI.

Emulating cameras via post-capture processing. The fine temporal granularity of single-photon acquisition makes it possible to emulate a wide variety of imaging modalities by performing computations *after the fact* [Sundar et al. 2023]. These imaging modalities include conventional grayscale cameras, exposure bracketing, high-speed video compressive systems, event cameras, and even cameras that (synthetically) move during exposure (see Fig. 2 (bottom row)). Our demonstration will allow users to emulate one or multiple such modalities in an interactive manner from live SPAD acquisition.

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